

```
%runfile D:/Farhana/neetu/neetu/evaluation.py --wdir
Loaded: (405894, 34)
100%|██████████| 405894/405894 [00:09<00:00, 41139.20it/s]
100%|██████████| 405894/405894 [00:09<00:00, 43006.65it/s]
100%|██████████| 405894/405894 [00:09<00:00, 42774.05it/s]
```

```
label_edge_cloud
0    317286
1     88608
Name: count, dtype: int64
```

```
Label distribution:
label_edge_cloud
0    317286
1     88608
Name: count, dtype: int64
```

```
Final cleaned dataframe info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 405894 entries, 0 to 405893
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   priority              405894 non-null  int32
1   scheduling_class      405894 non-null  int32
2   req_cpu               405894 non-null  float64
3   req_memory            405894 non-null  float64
4   avg_cpu_usage         405894 non-null  float64
5   max_cpu_usage         405894 non-null  float64
6   cluster               405894 non-null  int16
7   machine_id            405894 non-null  category
8   event                 405894 non-null  object
9   failed                405894 non-null  int8
10  label_edge_cloud      405894 non-null  int64
11  label_name            405894 non-null  object
dtypes: category(1), float64(4), int16(1), int32(2), int64(1), int8(1), object(2)
memory usage: 30.2+ MB
None
```

```
Sample:
   priority  scheduling_class  req_cpu  ...  failed  label_edge_cloud  label_name
0      200           3  0.020660  ...    1         0      cloud
1      360           2  0.007240  ...    1         1      edge
2      103           2  0.048584  ...    0         0      cloud
3      200           3  0.070435  ...    1         0      cloud
4         0           2  0.002449  ...    0         0      cloud
```

```
[5 rows x 12 columns]
Epoch 5/35, Loss: 0.9039
Epoch 10/35, Loss: 0.8350
Epoch 15/35, Loss: 0.8152
```

Epoch 20/35, Loss: 0.8102
Epoch 25/35, Loss: 0.8070
Epoch 30/35, Loss: 0.8028
Epoch 35/35, Loss: 0.8013

Training Time: 221.9726 seconds

Scheduling Tasks: 100%[██████████] 405894/405894 [18:35<00:00, 363.92it/s]

Feedback log preview:

	task_id	assigned_node	status	req_cpu	req_memory	priority
0	0	cloud_2	success	0.020660	0.014435	200
1	1	edge_1	success	0.007240	0.001303	360
2	2	cloud_1	success	0.048584	0.004166	103
3	3	cloud_1	success	0.070435	0.041626	200
4	4	cloud_1	success	0.002449	0.000232	0

Feedback learning applied.

	precision	recall	f1-score	support
Cloud	1.00	0.98	0.99	63513
Edge	0.93	0.99	0.96	17666
accuracy		0.98		81179
macro avg	0.96	0.99	0.97	81179
weighted avg	0.98	0.98	0.98	81179

Model Performance Metrics:

Accuracy : 98.2138

Precision: 92.8541

Recall : 99.4453

F1 Score : 96.0367

Average Scheduling Overhead: 0.000977 sec/task

===== Scalability Results =====

Tasks: 1000

Avg Latency : 0.003147

Success Rate : 48.94%

Tasks: 5000

Avg Latency : 0.003055

Success Rate : 48.94%

Tasks: 10000

Avg Latency : 0.002994

Success Rate : 48.94%

Tasks: 20000

Avg Latency : 0.003024

Success Rate : 48.94%

===== Load Balancing Results =====

Memory Load Variance : 151264.1263
Normalized CPU Load Variance : 0.0002
Balancing Accuracy: 79.08142719072166
Hotspots Detected : 0

===== TASK PERFORMANCE METRICS =====

Average Task Latency : 0.0181 units
Deadline Miss Rate : 0.00%
Average Waiting Time : 0.0126 units

===== Edge-Cloud Orchestration Accuracy =====

Edge/Cloud Offloading Accuracy: 98.21382377215782
Average Prediction Time per Task: 0.157383 ms
Baseline Cloud Offloads : 63513
Predicted Cloud Offloads: 62259
Reduction in Unnecessary Cloud Offloading: 1254
Reduction Percentage: 1.97%

===== Fault Tolerance and Reliability =====

Total Tasks : 405894
Successfully Assigned: 198656
Failed Assignments : 207238
Average Failover Attempts: 2.02
Total Critical Tasks : 333885
Failed Critical Tasks: 169611

===== Overall System Performance Summary =====

===== Additional Metrics =====

Average Latency Reduction: 20.14%
Task Completion Rate: 48.94%
Load Variance (CPU): 0.0002
Task Migration Success Rate: 100.00%
Orchestration Decision Accuracy: 48.11%

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